

Random Forest for Mechanical Engineering

A beginner-friendly tutorial on classification and regression

Prepared as a self-study and teaching resource for Mechanical Engineering students

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Preface

This tutorial is written for Mechanical Engineering students who are completely new to Artificial Intelligence (AI) and Machine Learning (ML). It begins with intuition, builds through decision trees and ensemble learning, and then develops the mathematics of Random Forest in a careful, step-by-step way. Throughout the document, the emphasis is on engineering understanding, model interpretation, and sound academic explanation rather than programming or software use.

Learning objectives

After studying this document, a reader should be able to:

- explain what AI, ML, supervised learning, classification, and regression mean;
- describe how a decision tree works and why a single tree can be unstable;
- explain how Random Forest combines many trees through bootstrap sampling, random feature selection, and aggregation;
- understand the central mathematics of Random Forest, including majority voting, averaging, variance reduction, the bias-variance tradeoff, out-of-bag error, and feature importance; and
- interpret how Random Forest can support Mechanical Engineering tasks such as material classification, material strength prediction, structural health monitoring, machined surface assessment, bearing failure susceptibility mapping, mechanics of materials and structures risk screening, and coolant quality prediction.

Detailed outline

1. Mounting bases in AI and ML

- Introduce AI as the broad field of intelligent systems and ML as the branch that learns patterns from data.
- Define supervised learning, input variables, target variables, classification, and regression.
- Explain why data-driven modelling matters in Mechanical Engineering.

2. Intuition behind Random Forest

- Build the central idea of combining many decision trees.
- Introduce ensemble learning in plain language.
- Explain why majority voting and averaging can produce more reliable predictions than a single tree.

3. Decision tree mounting bases

- Explain the structure of a decision tree, including nodes, branches, and leaves.
 - Show why a single tree can overfit and why it can change substantially when the training data change.
 - Motivate the need for diversity among trees.
4. **Random Forest for classification**
- Explain bootstrap sampling, random feature selection at each split, and majority voting.
 - Discuss binary and multiclass classification.
 - Explain how class votes should be interpreted.
5. **Random Forest for regression**
- Explain how the same forest idea is used when the target is numerical.
 - Show how averaging replaces majority voting.
 - Compare the classification and regression versions.
6. **Mathematical mounting bases**
- Introduce notation.
 - Define bootstrap sampling mathematically and conceptually.
 - Explain how classification trees and regression trees are grown.
 - Present formal aggregation rules for voting and averaging.
 - Explain why randomness reduces correlation among trees and why that helps generalization.
 - Introduce variance reduction, bias, variance, the bias-variance tradeoff, out-of-bag error, and feature importance.
7. **Mechanical Engineering applications**
- Present several application domains in which Random Forest has been used.
 - Emphasize the engineering meaning of inputs and outputs.
8. **Worked example: classification**
- Use a material classification example to show how several trees contribute to one final class decision.
 - Interpret the result in engineering terms.
9. **Worked example: regression**
- Use a mechanical material strength example to show how several trees contribute to one numerical prediction.
 - Interpret the result in engineering terms.
10. **Model understanding and practical interpretation**
- Summarize strengths, limitations, and situations where Random Forest performs well.
 - Compare Random Forest conceptually with simpler modelling approaches.
11. **Beginner support**

- Clarify training, testing, overfitting, underfitting, and generalization.
- Identify common mistakes beginners make.
- Explain how to decide whether a Mechanical Engineering problem is classification or regression.

12. Conclusion

- Reinforce the key conceptual and mathematical ideas and the difference between classification and regression.

1. Introduction to Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) is the broad field concerned with systems that perform tasks that normally require intelligent behaviour, such as recognising patterns, making decisions, or learning from experience (Russell & Norvig, 2021). Machine Learning (ML) is a major subfield of AI. In ML, a model is not written as a long list of hand-crafted rules. Instead, the model learns a mapping from data. In simple language, the model studies examples and then uses what it has learned to make predictions for new cases (Hastie et al., 2009; James et al., 2021).

A **data set** is a collection of observations. In Mechanical Engineering, an observation may be a material sample, a material sample, a machined surface segment, a mechanical component inspection record, a slope unit, or a coolant sample. Each observation contains **input variables**. These are measured quantities used to make a prediction. They are also called **features** or **predictor variables**. The quantity we want to predict is called the **target variable**, **response**, or, in classification problems, the **label**.

A **model** is a mathematical rule that maps the inputs to the target. In Mechanical Engineering, models may be based on mechanics, empirical equations, or data-driven learning. Machine learning does not replace mechanics, materials science, or code-based design. Instead, it complements them when relationships are complicated, nonlinear, noisy, site-specific, or difficult to express through a single closed-form equation (James et al., 2021; Sonbul & Rashid, 2023).

Table 1. Foundational vocabulary used throughout the tutorial.

Term	Simple meaning	Mechanical Engineering example
Observation	One case or record in the data set	One material sample from a test coupon
Input variable or feature	A measured quantity used to make a prediction	Liquid limit, moisture content, water-cement ratio
Target variable	The output the model tries to predict	Material class, compressive strength, damage state
Model	A mathematical mapping from inputs to outputs	A rule that predicts material strength from mix properties
Training	The stage where the model learns from known examples	Learning from mixes whose strength is already known
Prediction	Applying the model to a new case	Estimating the strength of a new mix
Generalization	Performing well on unseen data	Good predictions for future field or laboratory data

1.1 Supervised learning

Random Forest belongs to **supervised learning**. In supervised learning, each training observation includes both the input variables and the correct output. The model learns from these paired examples. If a material data set contains measured index properties together with the known material class, the task is supervised. If a material strength data set contains mixture proportions together with measured compressive strength, that is also supervised (James et al., 2021).

The supervised learning problem can be written conceptually as follows. Each observation has an input vector x and an output y . A learning algorithm uses a training data set

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$$

to estimate a rule $\hat{f}$ so that, for a new input x , the prediction is $\hat{y} = \hat{f}(x)$.

The hat on $\hat{f}$ indicates that the rule is estimated from data rather than known exactly. The quantity n is the number of observations in the training data set.

1.2 Classification and regression

Two of the most important supervised learning tasks are **classification** and **regression**.

Classification means that the target is a category. The model chooses among classes such as “stable” and “unstable”, “good” and “poor”, or several material or damage classes.

Regression means that the target is numerical. The model predicts a quantity such as compressive strength in megapascal, deflection in millimetre, or chloride concentration in milligram per litre.

Table 2. Classification and regression in Mechanical Engineering.

Question	Task type	Typical output
Is this material sample predominantly clay, silt, or sand?	Classification	A discrete class label
Is this machined surface segment in good, fair, or poor condition?	Classification	A discrete class label
What is the expected 28-day compressive strength of this material sample?	Regression	A numerical value
What is the predicted bearing capacity of this footing?	Regression	A numerical value

1.3 Where Random Forest fits

Random Forest is a **tree-based ensemble method** for supervised learning (Breiman, 2001). A **decision tree** is a model that makes predictions by asking a sequence of yes-no style questions about the input variables. An **ensemble** is a collection of models whose outputs are combined. Random Forest therefore belongs to the family of ensemble learning methods built from decision trees (Breiman, 1996, 2001).

1.4 Why these ideas matter in Mechanical Engineering

Mechanical Engineering data often have several characteristics that make Random Forest attractive. First, the relationship between inputs and outputs may be **nonlinear**, meaning that the output does not change proportionally with the input. Second, important **interactions** may exist, meaning that the effect of one variable depends on another. Third, measurements may be noisy, site-specific, or collected from heterogeneous materials and environments. Random Forest can handle such

complexity without requiring a single global equation to be guessed in advance (Catani et al., 2013; Liu et al., 2024; Xu et al., 2021; Zhang et al., 2019).

Section summary. AI is the broad field, ML is the learning-from-data part of AI, and supervised learning uses data sets where the correct outputs are known during training. Random Forest is a supervised, tree-based ensemble method that is useful when Mechanical Engineering relationships are complex, nonlinear, and data rich.

2. Intuition behind Random Forest

The central idea of Random Forest is easy to state. Instead of trusting a single decision tree, we build many trees and combine their predictions. If the task is classification, the trees vote. If the task is regression, the trees average their numerical predictions (Breiman, 2001).

This is an example of **ensemble learning**. In plain language, ensemble learning means combining several imperfect models so that the group performs better than a typical individual model. The logic is similar to asking several independent examiners to assess a machined surface segment instead of relying on only one examiner. If the examiners are reasonably skilled and do not all make the same mistake, the combined judgment is usually more reliable.

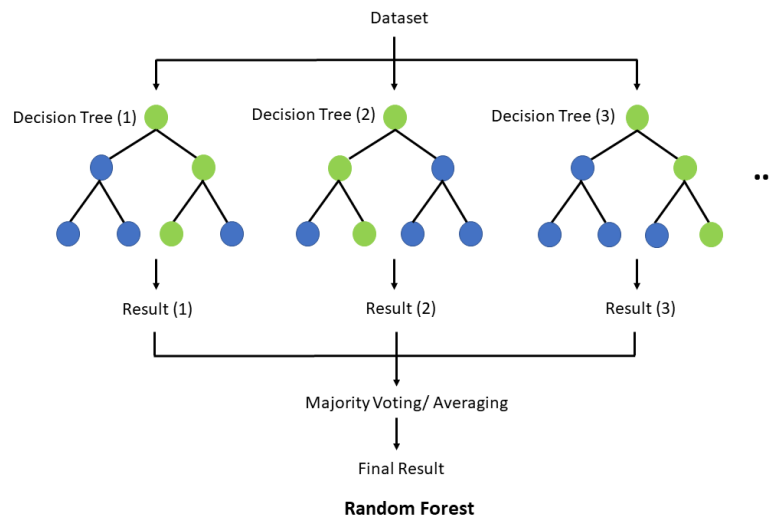


Figure 1. A conceptual diagram of Random Forest showing the repeated construction of many decision trees and the combination of their outputs. Source: TseKiChun (2021).

2.1 Why many trees can outperform one tree

A single tree can be accurate for the data used to build it, but it can also be unstable. A small change in the training data may produce a noticeably different tree. When many such trees are combined, individual errors can partially cancel. In classification, if several trees make different mistakes but most agree on the correct class, the majority vote can still be correct. In regression, if some trees predict slightly above the true value and others slightly below, averaging can reduce the overall fluctuation (Hastie et al., 2009; James et al., 2021).

The key word here is **diversity**. If all trees were identical, combining them would add little value. Random Forest therefore injects randomness in two places. First, each tree is trained on a different bootstrap sample of the data. Second, each split inside a tree considers only a random subset of the available features. Together, these steps make the trees different from one another, which is crucial for the success of the ensemble (Breiman, 2001).

2.2 Intuition before equations

Suppose eleven trees classify a machined surface segment. If seven trees say “poor”, three say “fair”, and one says “good”, the forest predicts “poor”. The vote proportions also convey useful information. A 7-3-1 split indicates less agreement than an 11-0-0 split.

Now suppose eleven trees predict material strength. If their predictions cluster around 35 megapascal, the average will also be near 35 megapascal and will typically be more stable than the prediction from any one tree. A forest therefore behaves like a smoothing mechanism over many tree-based decisions.

Section summary. Random Forest is powerful because it combines many diverse decision trees. The diversity comes from resampling the data and randomly restricting which features can compete at each split. Voting improves classification stability, and averaging improves regression stability.

3. Mounting bases from decision trees

A **decision tree** is a model that partitions the input space through a sequence of questions. Each internal point where a question is asked is called a **node**. Each answer creates a **branch**. A terminal point with a final prediction is called a **leaf**. For example, a mechanics of materials and structures tree may first ask whether plasticity index exceeds a threshold, then whether fines content exceeds another threshold, and then assign a material class at a leaf.

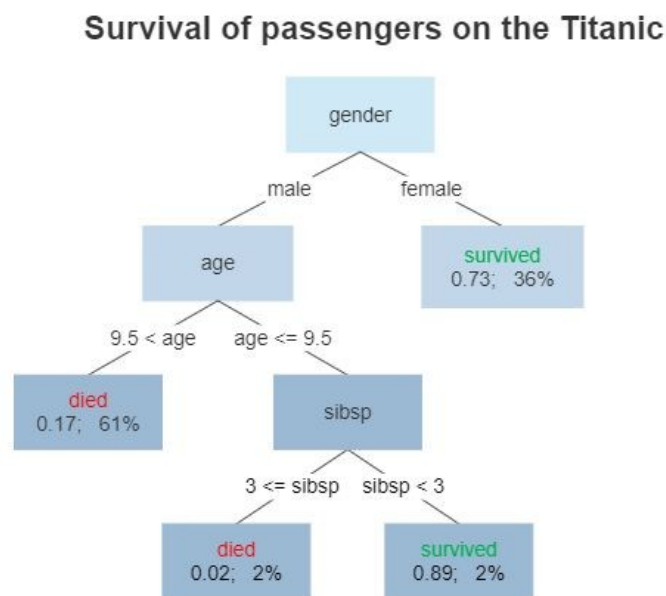


Figure 2. Example of a decision tree structure for a classification problem. The original illustration uses passenger survival data, but it is included here to show the general logic of recursive splitting and leaf-level prediction. Source: Gilgolds (2020).

3.1 Classification trees and regression trees

A **classification tree** predicts a category. It tries to create child nodes that are more homogeneous in class composition than the parent node. A **regression tree** predicts a number. It tries to create child nodes where the response values are more similar to each other than they were before the split (Hastie et al., 2009; James et al., 2021).

The tree-building process is called **recursive partitioning**. The word recursive means the same idea is applied repeatedly. The algorithm keeps splitting the data into smaller subsets until a stopping rule is reached.

3.2 Why a single tree can overfit

Overfitting occurs when a model learns not only the underlying pattern but also the accidental noise in the training data. A deep tree can keep splitting until very small groups are isolated. This can make the tree fit the training data extremely well, but the resulting pattern may not generalize to new observations (James et al., 2021).

Single trees are also **high-variance** models. In this context, high variance means that the fitted model can change a great deal when the training data change slightly. If a different sample of the same population were collected, the top split of the tree might change, and then many later splits could change as well (Hastie et al., 2009).

3.3 How Random Forest improves on a single tree

Random Forest retains the attractive parts of trees. Trees can represent nonlinear relationships, threshold effects, and interactions. At the same time, Random Forest reduces the instability of a single tree by averaging over many trees (Breiman, 2001).

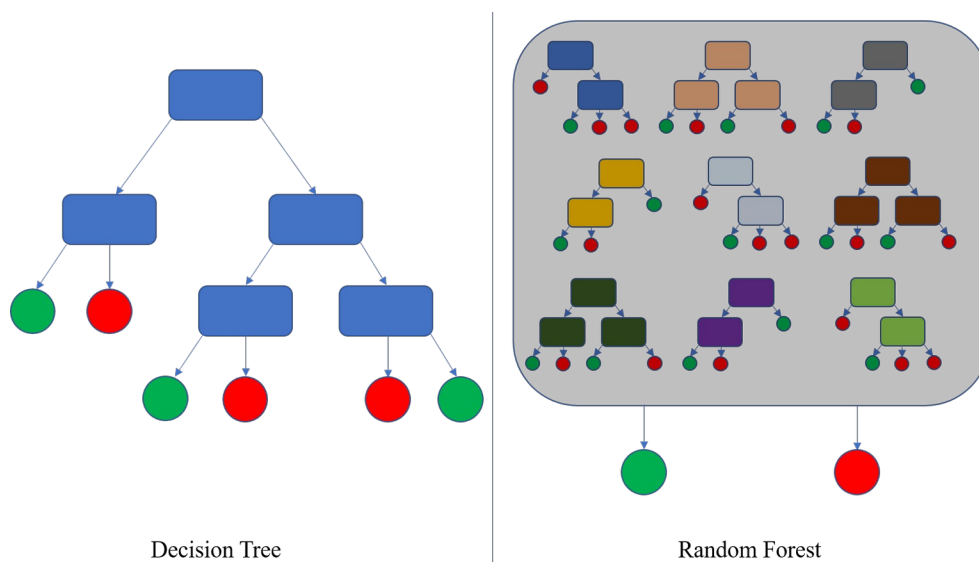


Figure 3. Conceptual comparison between a single decision tree and a Random Forest ensemble. The forest combines many tree predictions to improve stability. Source: Beauchamp (2020).

3.4 Why diversity matters

If every tree in the forest sees the same data and is allowed to examine exactly the same set of features at every split, the trees become too similar. Then averaging helps only a little. Diversity is important because averaging highly correlated trees is much less effective than averaging trees that make somewhat different errors. Random Forest uses bootstrap sampling and random feature selection to create this diversity systematically (Breiman, 2001).

Section summary. A decision tree is an intuitive model built from repeated splits, but a single tree can overfit and can be unstable. Random Forest builds many different trees and combines them, which preserves flexibility while reducing instability.

4. Random Forest for classification

In a Random Forest classification problem, the target variable is a class label. Let the classes be $1, 2, \dots, K$, where K is the number of classes. The forest is trained through the following conceptual steps (Breiman, 2001; James et al., 2021).

4.1 Step 1: draw a bootstrap sample for each tree

A **bootstrap sample** is created by sampling from the original training data **with replacement**. Sampling with replacement means that, after an observation is selected, it is returned to the pool before the next draw. Therefore the same observation may appear multiple times in one tree's training set, while some observations may be omitted entirely from that sample (Breiman, 1996, 2001).

4.2 Step 2: grow a classification tree on that sample

The tree is grown using the bootstrap sample rather than the full original data set. At each node, the algorithm does not examine all available features. Instead, it examines only a random subset of them. Among that subset, it chooses the split that best separates the classes. This creates variety from tree to tree, even if the overall data set is unchanged.

4.3 Step 3: repeat for many trees

The process is repeated many times to produce a forest of trees. Each tree has seen a different resampled version of the data and has considered different feature subsets during splitting.

4.4 Step 4: use majority voting for prediction

To classify a new observation, we send it down every tree. Each tree produces one class prediction. The Random Forest prediction is the class with the largest number of votes:

$$\hat{y}(x) = \arg \max_{k \in \{1, \dots, K\}} \sum_{b=1}^B 1\{T_b(x) = k\},$$

where B is the number of trees, $T_b(x)$ is the class predicted by tree b , and $1\{A\}$ is the **indicator function**, which equals 1 when statement A is true and 0 otherwise.

4.5 Binary and multiclass classification

If there are only two classes, the problem is **binary classification**. An example is “bearing failure” versus “no bearing failure”. If there are more than two classes, the problem is **multiclass classification**. An example is “good”, “fair”, and “poor” machined surface condition.

Majority voting works in both settings. In binary classification, the forest chooses the class with more than half the votes. In multiclass classification, the forest chooses whichever class receives the largest vote count, even if that count is not more than half of all votes.

4.6 Interpreting classification outcomes

The vote proportion for a class can be written as

$$\hat{P}(Y = k \mid x) = \frac{1}{B} \sum_{b=1}^B 1\{T_b(x) = k\}.$$

This quantity is often used as a probability-like score. If 80 out of 100 trees vote for one class, the estimated class proportion is 0.80. However, beginners should remember that vote proportions are not automatically perfect probabilities. They are best treated as useful measures of relative support unless additional calibration analysis has been carried out (Hastie et al., 2009; James et al., 2021).

Section summary. In classification, Random Forest trains each tree on a bootstrap sample, restricts the candidate features at each split, and then aggregates through majority voting. The final class is the most supported class across the ensemble.

5. Random Forest for regression

In a Random Forest regression problem, the target variable is numerical rather than categorical. The training logic is almost the same as in classification, but the tree-growing criterion and the final aggregation rule are different (Breiman, 2001).

5.1 Bootstrap samples and regression trees

Each tree is still trained on a bootstrap sample drawn with replacement from the original data. Each node still considers only a random subset of features as possible split candidates. The difference is that a regression tree seeks splits that make the response values within each child node as similar as possible. In practice, similarity is usually measured through squared error (Hastie et al., 2009; James et al., 2021).

5.2 Averaging predictions across trees

For a new input x , each tree produces a numerical prediction $T_b(x)$. The Random Forest regression prediction is the average:

$$\hat{y}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x).$$

This averaging is a major reason for the success of the method. Individual trees can fluctuate a great deal, but the average tends to be smoother and more stable.

5.3 Similarities and differences between classification and regression forests

The classification and regression versions of Random Forest share the same core principles: bootstrap resampling, random feature selection, many trees, and aggregation. The main differences are the type of target variable, the split criterion used during tree growth, and the way predictions are combined.

Table 3. Comparison of Random Forest classification and Random Forest regression.

Aspect	Classification forest	Regression forest
Target variable	Categorical	Numerical
Tree output	Class label	Number
Split objective	Increase class purity	Reduce response variability
Final aggregation	Majority vote	Arithmetic mean
Example	Material class, machined surface class	Material strength, bearing capacity

Section summary. Random Forest regression uses the same ensemble logic as Random Forest classification, but it predicts numbers and averages them. The forest therefore smooths the instability of individual regression trees.

6. Mathematical mounting bases

This section develops the mathematical core of Random Forest in a beginner-friendly but rigorous way. The aim is not to memorise formulas, but to understand what each formula is expressing.

6.1 Notation

We denote the training data set by

$$D = \{(x_i, y_i)\}_{i=1}^n,$$

where:

- n is the number of observations;
- $x_i = (x_{i1}, x_{i2}, \dots, x_{ip})^T$ is the vector of p input variables for observation i ; and
- y_i is the target for observation i .

The superscript T indicates transpose, meaning that the input variables are viewed as a column vector.

Table 4. Main symbols used in the mathematics of Random Forest.

Symbol	Meaning
n	Number of observations in the training data
p	Number of input variables
B	Number of trees in the forest
x_i	Input vector for observation i
y_i	Target value or class label for observation i
T_b	Tree b in the forest
m	Number of candidate features randomly offered at each split
K	Number of classes in a classification problem
$f(x)$	Forest prediction for input x

6.2 How an individual tree is constructed

A tree is built by repeatedly choosing a split of the form

$$x_j \leq s,$$

where x_j is one feature and s is a threshold value. The split divides the current node into a left child and a right child.

Classification split criterion

Suppose node t contains observations from K classes. Let $\hat{p}_{tk}$ be the proportion of observations in node t that belong to class k . A common impurity measure is the **Gini impurity**:

$$G(t) = 1 - \sum_{k=1}^K \hat{p}_{tk}^2.$$

This quantity is small when the node is nearly pure and larger when classes are mixed. A candidate split is good if it reduces impurity substantially. If the split sends the observations to child nodes t_L and t_R , the impurity reduction is

$$\Delta G = G(t) - \frac{n_{t_L}}{n_t} G(t_L) - \frac{n_{t_R}}{n_t} G(t_R),$$

where n_t is the number of observations in node t , and n_{t_L} and n_{t_R} are the numbers in the two child nodes. The tree chooses the candidate split that maximises ΔG among the feature-threshold pairs it is allowed to consider (Hastie et al., 2009; James et al., 2021).

Regression split criterion

For regression, a node is desirable when the response values inside it are similar. Let $\hat{y}_t$ be the mean response in node t . A common impurity measure is the **residual sum of squares** within the node:

$$RSS(t) = \sum_{i \in t} (y_i - \hat{y}_t)^2.$$

A good split minimises the weighted sum of child-node variability:

$$RSS(t_L) + RSS(t_R).$$

Thus, a regression tree chooses splits that create children with lower within-node numerical variability (Hastie et al., 2009; James et al., 2021).

6.3 Bootstrap sampling, conceptually and mathematically

For each tree b , Random Forest draws a bootstrap sample of size n from the original data set with replacement. Let the sampled indices be

$$I^{(b)} = (i_1^{(b)}, i_2^{(b)}, \dots, i_n^{(b)}),$$

where each index is drawn independently and uniformly from $\{1, 2, \dots, n\}$. The bootstrap training data for tree b are then

$$D^{(b)} = \{(x_{i_r^{(b)}}, y_{i_r^{(b)}})\}_{r=1}^n.$$

Because sampling is with replacement, some observations appear more than once in $D^{(b)}$ and some do not appear at all.

A useful fact is the probability that a particular observation is omitted from one bootstrap sample. The probability that observation i is not selected in a single draw is $1 - 1/n$. Since there are n draws, the probability that observation i is absent from the entire bootstrap sample is

$$\left(1 - \frac{1}{n}\right)^n.$$

As n becomes large, this approaches $e^{-1} \approx 0.368$, where e is the base of the natural logarithm.

Therefore, about 36.8% of the original observations are **out of bag** for a given tree, and about 63.2% appear at least once in that tree's bootstrap sample (Breiman, 1996, 2001).

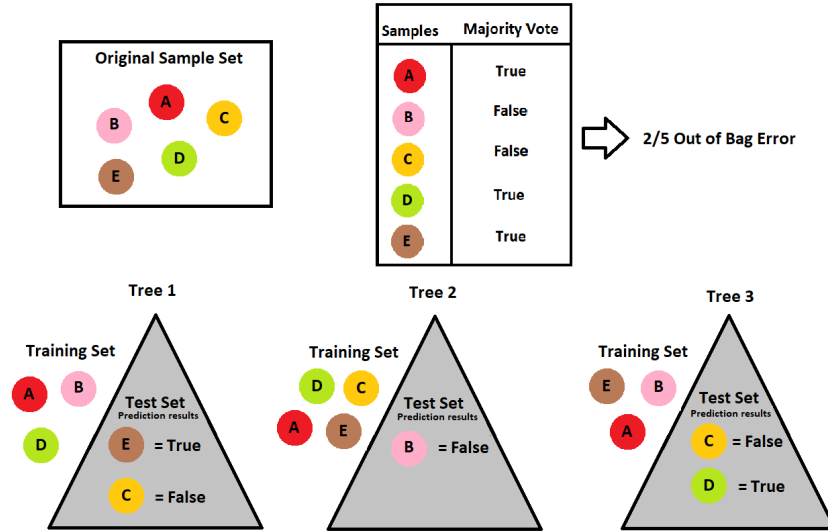


Figure 4. Illustration of the out-of-bag idea. Each tree is trained on a bootstrap sample, so some observations are left out for that tree and can be used as internal validation cases. Source: Analytics Vidhya (2020).

6.4 Random feature selection at each split

Suppose there are p total input variables. At each node, Random Forest randomly selects a subset $M_t \subseteq \{1, 2, \dots, p\}$ containing m candidate features. The split search is performed only within this subset. Symbolically,

$$|M_t| = m, m < p \text{ in the usual Random Forest setting.}$$

If one predictor is extremely strong, a standard tree would tend to choose it near the top of many trees, making the forest too similar. By limiting each node to a random subset of features, Random Forest forces some trees to explore alternative predictors and interactions. This lowers correlation among trees, which is essential for effective averaging (Breiman, 2001).

6.5 Aggregation across trees

Formal majority voting for classification

Let $T_b(x)$ be the class predicted by tree b for input x . The Random Forest classification rule is

$$\hat{f}(x) = \arg \max_{k \in \{1, \dots, K\}} \sum_{b=1}^B 1\{T_b(x) = k\}.$$

The forest therefore chooses the class with the largest vote count.

Formal averaging for regression

Let $T_b(x)$ be the numerical prediction of tree b for input x . The Random Forest regression rule is

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x).$$

The mean is the final prediction.

6.6 Why randomness improves generalization

Generalization means performing well on new, unseen data rather than only on the training data. A deep tree typically has low bias but high variance. In plain language, it can represent complicated patterns, but it is very sensitive to fluctuations in the training sample. Random Forest controls this

sensitivity by averaging many trees, and it makes averaging effective by reducing correlation among them through randomization (Breiman, 2001; Hastie et al., 2009).

To see why correlation matters, suppose each tree prediction at a fixed input has variance σ^2 and that the pairwise correlation between two trees is ρ . Then the variance of the average tree prediction is

$$\text{Var}\left(\frac{1}{B} \sum_{b=1}^B T_b(x)\right) = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2.$$

This formula shows two important ideas.

1. Increasing the number of trees B reduces the term $((1-\rho)/B) \sigma^2$.
2. Lowering the correlation ρ between trees is equally important. If the trees are highly correlated, averaging cannot reduce variance very much.

This is why Random Forest uses both many trees and deliberate randomness.

6.7 Bias, variance, and the bias-variance tradeoff

For regression, a standard decomposition at a fixed input x is

$$E\left[(Y - \hat{f}(x))^2\right] = (\text{Bias}[\hat{f}(x)])^2 + \text{Var}[\hat{f}(x)] + \sigma_\varepsilon^2,$$

where $Y = f(x) + \varepsilon$, $f(x)$ is the true underlying mean response, and $\sigma_\varepsilon^2 = \text{Var}(\varepsilon)$ is the irreducible noise variance. The three terms mean the following (Hastie et al., 2009; James et al., 2021).

- **Bias** is systematic error. A high-bias model is too rigid and misses real structure.
- **Variance** measures sensitivity to the training sample. A high-variance model changes a lot when the sample changes.
- **Irreducible noise** is variation in the response that no predictive model can remove completely.

A single deep tree often has low bias and high variance. Random Forest usually keeps bias reasonably low while substantially reducing variance through averaging. That is why it often generalises better than one tree.

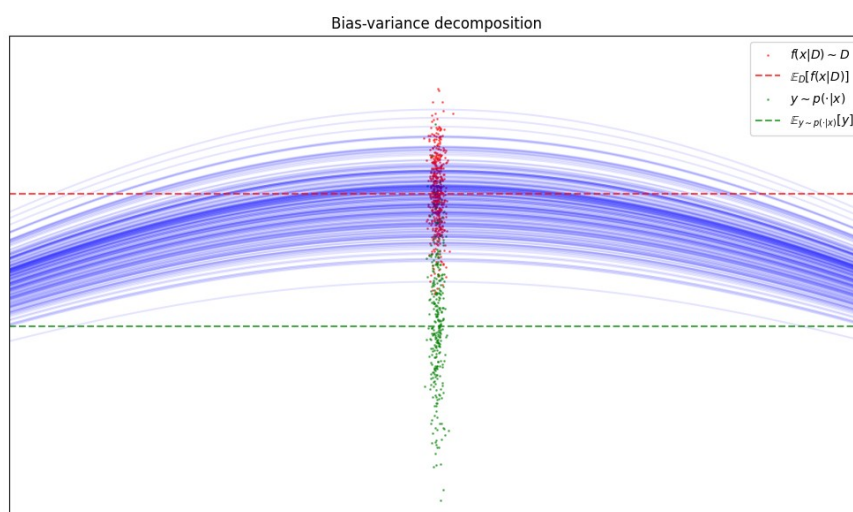


Figure 5. Bias and variance are two different sources of predictive error. Random Forest is often effective because it reduces variance while retaining much of the flexibility of a tree-based model. Source: Cosmia Nebula (2024).

6.8 Out-of-bag error

Because each tree leaves out roughly 36.8% of the observations, Random Forest provides a built-in internal validation mechanism. For observation i , let

$$B_i^{OOB} = \{b : i \notin D^{(b)}\}$$

be the set of trees for which observation i is out of bag.

For classification, the out-of-bag prediction is

$$\hat{y}_i^{OOB} = \arg \max_k \sum_{b \in B_i^{OOB}} 1\{T_b(x_i) = k\}.$$

The out-of-bag misclassification rate is then

$$Err_{OOB} = \frac{1}{n} \sum_{i=1}^n 1\{\hat{y}_i^{OOB} \neq y_i\}.$$

For regression, the out-of-bag prediction is the average over trees that did not use observation i for training:

$$\hat{y}_i^{OOB} = \frac{1}{|B_i^{OOB}|} \sum_{b \in B_i^{OOB}} T_b(x_i),$$

and a common out-of-bag error measure is the out-of-bag mean squared error,

$$MSE_{OOB} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i^{OOB})^2.$$

Out-of-bag error is useful because it estimates predictive performance without requiring a separate validation calculation for each tree. It should still be interpreted thoughtfully, but it is one of the practical strengths of Random Forest (Breiman, 2001).

6.9 Feature importance and its limitations

A common question in engineering is, “Which inputs matter most?” Random Forest can offer one answer through **feature importance** measures, but these measures must be interpreted carefully.

One popular approach is **mean decrease in impurity**. For feature j , it sums the impurity reductions over all nodes and all trees where feature j is used:

$$\mathfrak{I} p_{MDI}(j) = \frac{1}{B} \sum_{b=1}^B \sum_{t: v(t)=j} p(t) \Delta I_t,$$

where $v(t) = j$ means that node t splits on feature j , $p(t)$ is the proportion of observations reaching node t , and ΔI_t is the impurity reduction produced by that split.

Another approach is **permutation importance**. After the model is built, the values of one feature are randomly permuted while the others are left unchanged. If predictive performance deteriorates strongly, that feature was important. In symbols,

$$PI(j) = Err_{OOB}^{perm(j)} - Err_{OOB},$$

where $Err_{OOB}^{perm(j)}$ is the out-of-bag error after permuting feature j (Breiman, 2001; Fisher et al., 2019).

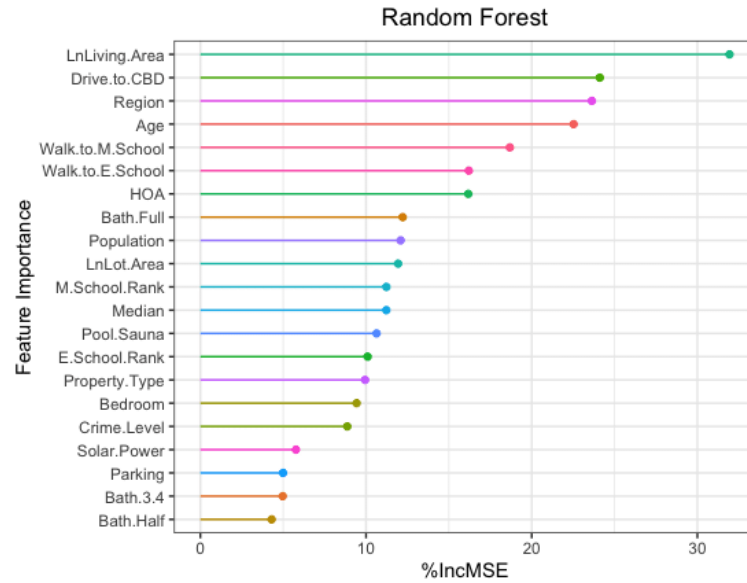


Figure 6. Illustrative feature-importance ranking from a Random Forest model in a different application area. The purpose here is to show the idea of ranking variables by their estimated predictive contribution, not to suggest that these variables are relevant to Mechanical Engineering. Source: Yazdani (2021).

Feature importance is helpful, but it is not the same as physical causation. A variable can appear important because it is strongly associated with the outcome, because it stands in for another correlated variable, or because of the way the data were collected. Impurity-based importance can also favour variables with many possible split points or many categories, and correlated predictors can make importance scores difficult to interpret (Fisher et al., 2019; Strobl et al., 2007).

Section summary. The mathematics of Random Forest combines tree splitting, bootstrap resampling, random feature subsets, and ensemble aggregation. The method often improves generalization because it reduces variance while maintaining flexibility. Out-of-bag error gives internal validation, and feature importance can be informative, but it has important interpretation limits.

7. Mechanical Engineering applications

Random Forest has been applied across many Mechanical Engineering subfields because it can capture nonlinear relationships, interactions, and mixed patterns in measured data. The method is especially attractive when the goal is prediction or screening and when the available information is multivariate and partly noisy.

Table 5. Representative Mechanical Engineering applications of Random Forest.

Application area	Typical task	Example inputs	Typical output	Representative source
Material classification	Classification	material characterizations, grain-size fractions, moisture content, density	Material class	Liu et al. (2024)
Material properties	Regression	Cement content, water-binder ratio, supplementary cementitious materials, age	Compressive strength	Zhang et al. (2019)
Machined surface assessment	Classification	Distress indicators, production throughput, climate, surface condition variables	Condition class or damage location	Guo & Hao (2021)
Structural health	Classification or regression	Vibration features, strain, acceleration, temperature	Damage state or anomaly	Zhou et al. (2013); Sonbul & Rashid

Application area	Typical task	Example inputs	Typical output	Representative source
monitoring			indicator	(2023)
Bearing failure susceptibility	Classification	Slope angle, lithology, flow-rate, land cover, elevation	Susceptibility class	Catani et al. (2013)
Mechanics of Materials and Structures design support	Regression	Standard Penetration Test values, density, footing dimensions, layer properties	Bearing capacity	Dutta et al. (2019)
Coolant quality assessment	Classification or regression	Dissolved oxygen, particulate concentration, nutrients, temperature, pH	Coolant quality class or indicator level	Xu et al. (2021)

7.1 Material classification

Material classification involves threshold effects, mixed variables, and overlapping class boundaries. Laboratory index properties may interact in ways that are difficult to summarise with a single linear equation. Random Forest can therefore be useful for preliminary material classification, screening, or support to more traditional mechanics of materials and structures interpretation, especially when many past labelled samples are available (Liu et al., 2024).

7.2 Prediction of mechanical material strength

Material strength depends on interacting material proportions, curing conditions, admixtures, and age. These relationships are often nonlinear. Random Forest regression can learn such patterns from experimental data and can be used to estimate compressive strength or other performance measures while still requiring engineering judgment and laboratory validation (Zhang et al., 2019).

7.3 Machined surface condition classification

Rotating machinery performance depends on duty-cycle loading, climate, material condition, maintenance history, and surface distress measures. Random Forest can support tasks such as condition classification and potential damage localization because it handles interaction-rich data and can rank contributing variables (Guo & Hao, 2021).

7.4 Structural health monitoring

Structural Health Monitoring (SHM) uses sensor data such as vibration responses, strains, or accelerations to infer structural state. These data are often high dimensional and noisy. Random Forest has been used for damage detection and classification because it is robust and can integrate many features without requiring a rigid functional form (Sonbul & Rashid, 2023; Zhou et al., 2013).

7.5 Bearing failure susceptibility assessment

Bearing failure susceptibility is influenced by topography, flow-rate, geology, land cover, and coolant-return conditions. Random Forest has been applied to this problem because it can model complex interactions among environmental variables and produce maps or class labels indicating relative susceptibility (Catani et al., 2013).

7.6 Mechanics of Materials and Structures risk identification and bearing capacity estimation

Random Forest can support mechanics of materials and structures screening problems where the output is either a category, such as “high” or “low” risk, or a numerical quantity, such as bearing capacity. Such applications benefit from the method’s ability to model nonlinear relations between site variables and engineering outcomes (Dutta et al., 2019).

7.7 Coolant quality prediction and classification

Thermal-fluid systems respond to thermal-fluid, chemical, and operational factors in complicated ways. Random Forest has been applied to classify coolant quality status or predict continuous indicators because it can flexibly incorporate multiple water-quality measurements and environmental descriptors (Xu et al., 2021).

Section summary. In Mechanical Engineering, Random Forest is useful in tasks where many measured variables jointly influence an outcome and where relationships may be nonlinear, interaction-rich, and noisy. It is particularly valuable for prediction, screening, and condition assessment.

8. Worked example 1: a Mechanical Engineering classification problem

This worked example uses a simplified material classification task. The example is intentionally conceptual. Its purpose is to show how a Random Forest combines several trees into one final class decision.

8.1 Problem context

Suppose an engineer has a data set of previously tested materials, each labelled according to the Unified Material Classification System (USCS). The goal is to use measured index properties to predict whether a new sample is most similar to:

- **ML**, low-plasticity silt;
- **CL**, lean clay; or
- **CH**, highly plastic clay.

In practice, formal material classification still depends on standard mechanics of materials and structures procedures. The model here is a data-driven aid for screening and interpretation, not a substitute for laboratory standards.

8.2 Inputs for the new sample

Assume the new sample has the following measured properties:

- liquid limit ω_L 42;
- plasticity index ω_P 19;
- fines content ω_F 58 %;
- sand fraction ω_S 24 %; and
- natural moisture content ω_N 21 %.

Five trees in the forest evaluate the sample. Because the trees were built from different bootstrap samples and different random feature subsets, they do not all follow the same path.

Table 6. Example votes from five classification trees for one material sample.

Tree	Dominant logic of the path	Tree prediction
1	High plasticity index and high fines content	CL
2	Liquid limit above threshold and moderate clay-like pattern	CL
3	Moisture and fines pattern closer to silt boundary	ML
4	Plasticity-driven split favours clay behaviour	CL
5	Combined effect of liquid limit and fines favours clay	CL

The vote count is therefore:

- CL: 4 votes
- ML: 1 vote
- CH: 0 votes

The forest prediction is the majority class, namely **CL**. The corresponding vote proportions are

$$\hat{P}(CL \mid x) = \frac{4}{5} = 0.80, \hat{P}(ML \mid x) = \frac{1}{5} = 0.20, \hat{P}(CH \mid x) = 0.$$

8.3 Engineering interpretation

The forest suggests that the sample behaves most like a lean clay rather than a silt or a highly plastic clay. The one dissenting tree is informative: it indicates that the sample may be near a class boundary. That means the prediction should be interpreted as strong evidence in favour of CL, but not as absolute certainty.

From an engineering perspective, this kind of model could help with preliminary site screening, checking consistency across large historical databases, or prioritising samples for further laboratory review. It should not replace index testing standards, geological understanding, or mechanics of materials and structures judgment.

Section summary. In classification, each tree casts one vote. The final class is the most common vote across the forest. The distribution of votes is also useful because it conveys how strongly the ensemble agrees.

9. Worked example 2: a Mechanical Engineering regression problem

This worked example uses a simplified mechanical material strength prediction task. Again, the goal is conceptual understanding rather than software calculation.

9.1 Problem context

Suppose a data set contains many material samples with known 28-day compressive strength. The input variables include material proportions and curing information. A Random Forest regression model is trained to estimate strength for new mixes.

Consider a new mix with the following properties:

- cement content 380 kg/m³;
- water-binder ratio 0.44;
- fly ash content 60 kg/m³;
- coarse aggregate content 1030 kg/m³; and
- curing age 28 days.

9.2 Predictions from individual trees

Five trees in the forest produce the following strength estimates.

Table 7. Example predictions from five regression trees for one material sample.

Tree	Main pattern emphasised by the tree	Predicted strength (MPa)
1	Water-binder ratio and age	34.2
2	Cement plus fly ash interaction	35.8
3	Aggregate content and ratio effects	33.6
4	Stronger influence of cement content	36.1
5	Balanced influence of all variables	34.7

The Random Forest prediction is the arithmetic mean:

$$\hat{y}(x) = \frac{34.2 + 35.8 + 33.6 + 36.1 + 34.7}{5} = 34.88 \text{ MPa}.$$

Rounded suitably, the forest predicts about **34.9 MPa**.

9.3 Engineering interpretation

If the design requirement were 30 MPa at 28 days, the prediction would suggest that the mix is likely to meet that target. However, the model prediction is not a replacement for compressive strength testing, code compliance, or quality-control procedures. It is a predictive aid.

This example also shows why averaging helps. No single tree perfectly represents the relationship. One tree predicts 33.6 MPa, another predicts 36.1 MPa, and the forest smooths these fluctuations into a more stable estimate.

Section summary. In regression, each tree produces a number, and the forest averages those numbers. The average is typically more stable than an individual tree prediction and is often easier to trust as a predictive summary.

10. Model understanding and practical interpretation

Random Forest is widely used because it often gives strong predictive performance with comparatively modest model assumptions. At the same time, it is not a universal solution.

10.1 Strengths of Random Forest

- It handles nonlinear relationships naturally.
- It can capture interactions among variables without requiring the analyst to specify all interactions in advance.

- It usually reduces the instability of a single decision tree.
- It works well for both classification and regression.
- It can provide internal validation through out-of-bag error.
- It can provide feature-importance measures that help exploratory interpretation (Breiman, 2001; James et al., 2021).

10.2 Limitations of Random Forest

- It is less interpretable than a single decision tree because the final prediction depends on many trees rather than one explicit rule set.
- It can require substantial computational effort for very large data sets or very large forests.
- It still requires thoughtful choices about the number of trees, the number of candidate features at each split, and the stopping behaviour of the trees.
- It can struggle with severe class imbalance if rare classes are not handled carefully.
- Feature-importance measures can be biased or misleading, especially when predictors are correlated or when some predictors have many possible split points (Fisher et al., 2019; Strobl et al., 2007).

10.3 Situations where Random Forest often performs well

Random Forest often performs well when:

- the predictor-response relationship is nonlinear;
- several predictors interact;
- the data contain noise but enough observations are available to average over instability;
- predictive accuracy is important; and
- a flexible baseline model is needed before more specialised methods are considered.

10.4 Comparison with simpler modelling approaches

Table 8. Conceptual comparison of several common modelling approaches.

Method	Main idea	Major advantage	Main limitation
Linear regression	Fit one global linear relation for a numerical target	High interpretability	Limited for complex nonlinear patterns
Logistic regression	Fit a probabilistic linear boundary for a categorical target	Clear coefficients and interpretation	May miss threshold effects and complex interactions
Single decision tree	Use recursive if-then rules	Easy to explain visually	Can overfit and be unstable
Random Forest	Combine many randomized trees	Strong predictive stability and flexibility	Less transparent than a single tree

The important point is not that Random Forest is always superior. If the true pattern is close to linear and interpretability is the main objective, a simpler model may be preferable. Random Forest is especially attractive when the relationship is complex and prediction is a central goal.

Section summary. Random Forest is powerful because it combines flexibility with variance reduction, but it trades some interpretability for predictive stability. Good engineering use requires understanding both its strengths and its limits.

11. Beginner support

This section addresses common conceptual difficulties that students encounter when first learning Random Forest.

11.1 Common mistakes beginners make

Table 9. Common beginner mistakes and better ways to think about them.

Common mistake	Better interpretation
Confusing the input variables with the target variable	First identify what is known before prediction and what must be predicted
Treating a numerical outcome as classification or a categorical outcome as regression	Ask whether the output is a class label or a number
Judging success only by training performance	Good models must also perform well on unseen data
Assuming vote proportion equals perfect probability	Vote share indicates support, but not guaranteed calibration
Treating important features as proven causes	Importance reflects predictive usefulness, not necessarily physical causation
Using the model as a replacement for engineering judgment	Models support decisions; they do not eliminate the need for domain knowledge

11.2 Training, testing, overfitting, underfitting, and generalization

The **training set** is the portion of data used to build the model. A **test set** is a separate set of observations used to evaluate how well the model performs on unseen cases. A model that performs extremely well on training data but poorly on unseen data is overfitting. A model that performs poorly even on the training data may be **underfitting**, meaning it is too simple to capture the essential pattern (James et al., 2021).

Generalization means that the model continues to perform well when confronted with new, comparable data. This is the real goal of predictive modelling. Random Forest often improves generalization relative to a single decision tree because it averages over many unstable trees rather than trusting one tree completely (Breiman, 2001).

11.3 How to decide whether a problem is classification or regression

A practical rule is simple:

- if the desired output is a category, it is a classification problem;
- if the desired output is a number, it is a regression problem.

Examples:

- “Is this mechanical component damaged or undamaged?” is classification.
- “What is the expected chloride concentration?” is regression.

- “Which material class is most likely?” is classification.
- “What is the expected compressive strength?” is regression.

11.4 Why Random Forest is powerful, but still requires thoughtful interpretation

Random Forest is often strong in practice because it makes relatively few rigid assumptions about functional form, it learns from multiple interacting inputs, and it reduces tree instability through aggregation. Yet good engineering interpretation still requires asking important questions:

- Are the inputs measured reliably?
- Does the training data represent the cases to which the model will be applied?
- Is the target variable well defined?
- Is the prediction being used for screening, design support, monitoring, or final decision-making?
- Do the model outputs align with known engineering principles and physical plausibility?

A model can be statistically impressive and still be poorly suited to a particular engineering decision if the data are unrepresentative or the interpretation is careless.

Section summary. Beginners learn Random Forest more effectively when they keep the modelling purpose clear, distinguish inputs from targets, evaluate performance on unseen data, and remember that predictive usefulness does not remove the need for engineering judgment.

12. Conclusion

Random Forest is one of the most important and widely used methods in supervised machine learning. It is built from a simple but powerful idea: combine many decision trees, each trained on a different bootstrap sample and each encouraged to explore different features, then aggregate their predictions. For classification, the forest uses majority voting. For regression, it uses averaging.

For Mechanical Engineering students, Random Forest is especially valuable because many engineering problems involve nonlinear behaviour, interacting variables, noisy observations, and complex material or environmental effects. The method can therefore support tasks such as material classification, material strength prediction, structural health monitoring, machined surface condition assessment, bearing failure susceptibility analysis, mechanics of materials and structures prediction, and coolant quality modelling.

The most important conceptual distinction to retain is this: classification predicts categories, while regression predicts numbers. The most important mathematical ideas to retain are bootstrap sampling, random feature selection, aggregation across trees, variance reduction, and the bias-variance tradeoff. With these ideas in place, a student has a strong basis for understanding why Random Forest often performs well in practice and how it should be interpreted responsibly in engineering applications.

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